

HAT-P-36b

R-0939

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_230316 · created 2026-07-29T12:39:46+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460019.7493 +/- 0.005 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.045
Transit depth (Rp/Rs)^2	2.47 +/- 1.4 [%]
Orbital Inclination (inc)	89.01 +/- 2.78
Airmass coefficient 1 (a1)	1.07 +/- 0.13
Airmass coefficient 2 (a2)	-0.085 +/- 0.096
Scatter in the residuals of the lightcurve fit is	12.44 %
Best Comparison Star	#2 - [451, 86]
Optimal Aperture	3.0
Optimal Annulus	3.0
Transit Duration (day)	0.0985 +/- 0.0052

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.157 ± 0.045 vs 0.126 (deviation 0.69σ)
Duration fit vs published	0.0985 d vs 0.0925 d (deviation 1.15σ)
Mid-time O–C	-12.3 min
Depth SNR	3.5
Residual scatter	12.44 %

Light curve

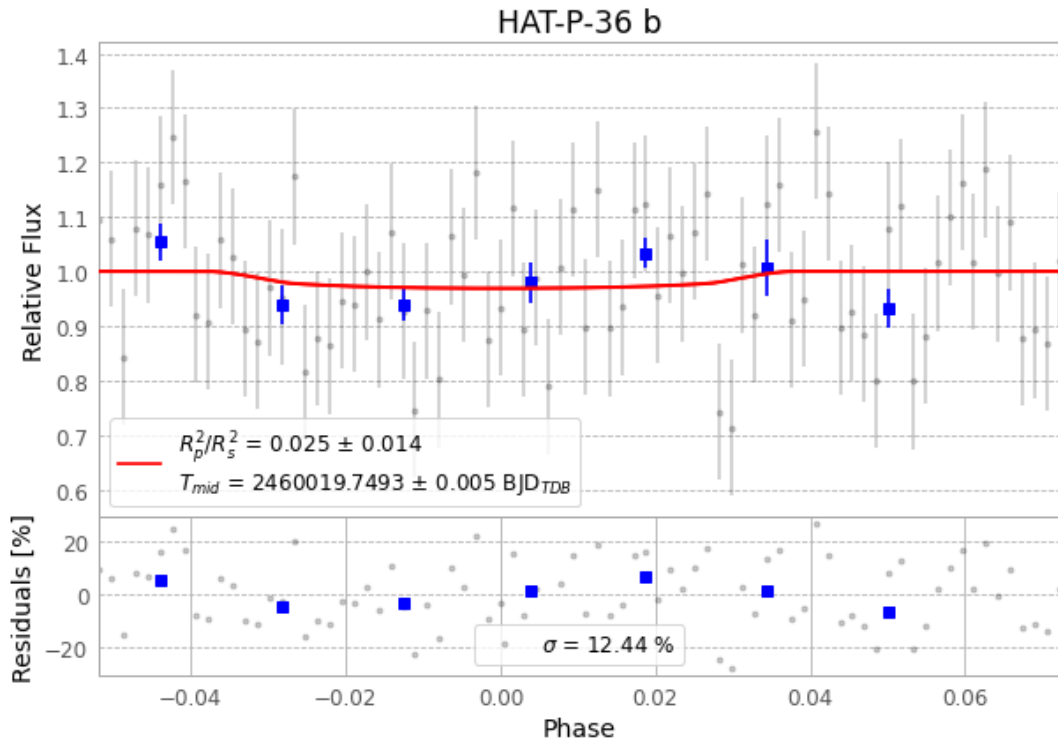


Figure 1 — FinalLightCurve_HAT-P-36 b_2023-03-15.png (SHA-256 5e22555deb7fafd8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [427, 454], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 be40b563974bb200...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

80 FITS frames (53 MB), HATP-36230316041215.FITS → HATP-36230316080915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-230316.log (09ba35c621da...), FinalLightCurve_HAT-P-36 b_2023-03-15.png (5e22555deb7f...), FinalLightCurve_HAT-P-36 b_2023-03-15.csv (bda36f0eb766...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 12:39Z.